
Question: - 1

Ana likes many activities. She likes acrobatics, alchemy, archery, art, Arabic dances, and many more. She joined a club that offers several classes. Each class has a time interval in every week. Ana wants to sign up for many classes, but since they overlap in time, she looks for a subset of non-overlapping classes to attend. A subset is non-overlapping if it does not contain two classes that overlap in time. If a class starts at the time another class ends, this is not considered overlapping. Ana decided to list all the non-overlapping non-empty subsets of classes. Then she will choose the subset she likes best. In order to predict the amount of paper needed to write the list, she wants you to calculate how many of these subsets there are.

Input

Each test case is described using several lines. The first line contains an integer N indicating the number of classes the club offers ($1 \leq N \leq 10^5$). Each of the next N lines describes a class using two integers S and E that represent the starting and ending times of the class, respectively ($1 \leq S < E \leq 10^9$). The end of input is indicated with a line containing a single -1.

Output

For each test case, output a single line with a single integer representing the number of non-overlapping non-empty subsets of classes. To make your life easier, output only the last 8 digits of the result. If the result has less than 8 digits, write it with leading zeros to complete 8 digits.

Example:

Input:

```
5
1 3
3 5
5 7
2 4
4 6
3
500000000 1000000000
1 500000000
1 500000000
1
999999999 1000000000
-1
```

Output:

```
00000012
00000005
00000001
```

Question: - 2

Aly being one of the smartest guys in his third grade class solved the question "How many ways to assign K cells for rooks in an $N \times M$ such that no two rooks are attacking each other" (A rook is a chess piece that can attack other pieces on the same row or the same column) but he was only able to solve it when $N \times M$ didn't exceed 10. After some days one of his classmates said he was able to solve it even if $N \times M$ reached 100. Aly kept asking how he did it and after questioning a lot of people he knew that he used a program to calculate this for him so he decided to challenge him in front of all his classmates but to do that he also needed a program that not only solve the problem when $N \times M$ reaches 100 but when N and M each reach 1,000,000 and he came for you to do that program for him but as he knows that the answer can be really big he wanted the program to output the number of ways modulo 1,000,000,007.

Input

The first and only line of input contains three numbers N, M and K
($1 \leq N, M \leq 1,000,000, 1 \leq K \leq N \times M$).

Output

The number of ways required modulo 1,000,000,007.

Example:

Input:

4 4 4

Output:

24

Question: - 3

A beehive is an enclosed structure in which some honey bee species live and raise their young. In this problem we consider a two-dimensional sketch of the beehives. Each beehive is composed of a certain number of cells, where each cell is a regular hexagon. Each cell may have some neighbors, which are other cells that share a side with that cell. A cell with exactly 6 neighbors is an internal cell, while a cell with fewer neighbors is an external one. Notice that an external cell can always be changed to internal by adding some neighbor cells.

We are interested in a particular class of beehives. This class of valid beehives is defined recursively as follows: a) a single cell is a valid beehive; and b) given a valid beehive B, if we add the minimum number of cells such that each external cell of B becomes an internal cell, the result is a valid beehive.

The number of cells in a valid beehive is called a beehive number. Given an integer N, you must decide whether it is a beehive number.

Input

Each test case is described using a single line. The line contains an integer N ($1 \leq N \leq 10^9$). The end of input is indicated with a line containing a single -1.

Output

For each test case, output a single line containing an uppercase “Y” if N is a beehive number, or an uppercase “N” otherwise.

Example:

Input:

```
43
1
7
19
15
-1
```

Output:

```
N
Y
Y
Y
N
```

Question: - 5

We are given a sequence S of N distinct integers. Denote by $S[i]$ as its i th element of S .

Hardik picks up a random permutation of S , breaks it into product of disjoint cycles & looks at cycle containing $S[i]$. He notes down the sum of all element of this cycle. Call the expected value of this sum as $\text{cycleSum}[i]$. Your task is to find the minimum value amongst all cycleSums .

Assume all permutations of these N numbers are equally likely.

Input

First line contains an integer T which denotes the number of test cases. Then follow description of T test scenarios. Each test scenario takes 2 lines. First line contains a single integer N , the size of S . Then follows second line containing N elements of S .

Output

Print answer for each test case, rounded to exactly one decimal place, in one line each.

Example:

Input:

```
2
1
1
2
1 2
```

Output:

```
1.0
2.0
```